

EGT214 Deep Learning Project

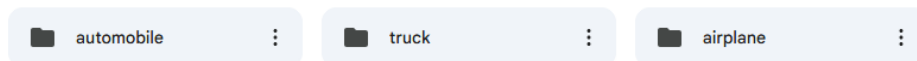
Objective:

To develop deep learning models to solve a real-world problem with the given dataset, and to apply the relevant techniques to evaluate and optimize the deep learning model and efficiency.

- Develop deep learning models for the given dataset and problem definition.
- Apply model tuning and optimization to improve the performance of deep learning models.
- Evaluate deep learning models using appropriate metrics for efficiency and effectiveness.
- Compare the different deep learning models on their performance with respect to the given dataset and problem definition.

Datasets (dataset_transport):

- Download dataset_transport from Brightspace
- Classes: 3 (airplane, automobile, truck)
- Train data: 6000 images
- Test data: 1500 images



Project Tasks:

1. Develop **TWO** deep learning models (**CNN** and **Transferred Learning**) for the classification task with the given dataset.
2. Develop a single Colab Notebook with below sections:
 - Data preparation
 - Build CNN model
 - Summary of CNN model development (<150 words): to describe your key considerations on design strategy, overfitting techniques, evaluation & optimization.
 - Build Transfer Learning model
 - Summary of Transfer Learning model development (<150 words): to describe your key considerations on pre-trained model selection, tuning strategies, overfitting techniques, evaluation & optimization.
 - Comparison of the 2 models (CNN and Transfer Learning) (<150 words): to compare and explain their results and performance, pros/cons, suitability for the given task.
 - Keep the running results within the notebook.
 - Add your adminNo as the suffix of model name (e.g. CNN_model_223456A).

Submission:

- Submit Colab notebook ([admin_project.ipynb](#)) via *EGT214 Project Submission (50%)* in Brightspace.
- **Deadline: By end of Week 18** (23:59 on 16/02/2024).

*This is an individual project. **ZERO** mark will be given if fully copy is found!

Assessment Rubrics

Marks will be allocated based on the task completeness, model development, techniques, performance, and the summary and comparison:

Items	Need Improvement	Satisfactory	Excellent
CNN Model (30%)	Poor development: • CNN architecture • regularization and augmentation • evaluation and optimization • performance and results	Satisfactory development: • CNN architecture • regularization and augmentation • evaluation and optimization • performance and results	Excellent development: • CNN architecture • regularization and augmentation • evaluation and optimization • Performance and results
Summary of CNN model (10%)	Insufficient and unclear description and explanation: • CNN design strategy • overfitting techniques - evaluation and optimization	Insufficient and clear description and explanation: • CNN design strategy • overfitting techniques - evaluation and optimization	Sufficient and clear description and explanation: • CNN design strategy • overfitting techniques - evaluation and optimization
Transfer Learning Model (30%)	Poor development: • pre-trained model selection • feature extraction and fine-tuning strategies • overfitting techniques • evaluation and optimization • performance and results • effort on customization of base model to accommodate training task	Satisfactory development: • pre-trained model selection • Feature extraction and fine-tuning strategies • overfitting techniques • Evaluation and optimization • Performance and results • effort on customization of base model to accommodate training task	Excellent development: • pre-trained model selection • Feature extraction and fine-tuning strategies • overfitting techniques • Evaluation and optimization • Performance and results • effort on customization of base model to accommodate training task
Summary of Transferred Learning model (10%)	Insufficient and unclear description and explanation: • pre-trained model selection • feature extraction and fine-tuning strategies • overfitting techniques • evaluation and optimization	Insufficient and clear description and explanation: • pre-trained model selection • feature extraction and fine-tuning strategies • overfitting techniques • evaluation and optimization	Sufficient and clear description and explanation: • pre-trained model selection • feature extraction and fine-tuning strategies • overfitting techniques • evaluation and optimization
Comparison of 2 models (10%)	Comparisons are not clear: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • understanding of problem	Comparisons are clear but could have done better: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • understanding of problem	Comparisons are clearly organized and conveyed: • results and performance • pros/cons • reasoning for decisions • comprehensive views and supporting evidence • understanding of problem
Execution Performance (10%)	Poor execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity	Satisfactory execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity	Excellent execution performance: • errors in notebook • execute correctly • notebook developed in sequences • originality & creativity